

Runyu Lu

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RESEARCH VISION

Design and Build Fast, Efficient, and User-friendly Infra for GenAI workloads(Robotics, LLM, Diffusion, AI Agent, etc).

EDUCATION

PhD of Computer Science Sept. 2024 - Present
University of Michigan, Research Areas: Machine Learning System. Advisors: Mosharaf Chowdhury and Ang Chen

Bachelor of Computer Science Sept. 2020 - June. 2024
Huazhong University of Science and Technology, Computer Excellence Program, GPA: 3.95 / 4.00

PUBLICATIONS AND ONGOING PAPERS

MuxServe: Flexible Multiplexing for Efficient Multiple LLM Serving **ICML 2024**
Jiangfei Duan, **Runyu Lu**, Haojie Duanmu, Xiuhong Li, Xingcheng ZHANG, Dahua Lin, Ion Stoica, Hao Zhang
Keyword: LLM Serving, GPU Streaming Multiprocessor, LLM Model Colocation

White-box Compiler Fuzzing Empowered by Large Language Models **OOPSLA 2024**
Chenyuan Yang, Yinlin Deng, **Runyu Lu**, Jiayi Yao, Jiawei Liu, Reyhaneh Jabbarvand, Lingming Zhang
Keyword: LLM for Compiler, LLVM, LLVM IR Fuzzing

INDUSTRIAL EXPERIENCE

ColossalAI: Design and Implement LLM/Diffusion Inference Engine *HPC-AI Tech*, March.2024 - July.2024
Mentors: Yuanheng Zhao, Shenggui Li, Hongxin Liu
Refactor and Design Diffusion/LLM Inference Engine for ColossalAI 🏆, 37k+ stars in Github.

LLVM: Optimize GPU Compiler Backend *SenseTime*, April 2023 - Sept.2023
4000+ line LLVM GPU Backend Optimization Codes, new backend under LLVM 🏆
GPU Compiler Optimization: Instruction Selection, Instruction Pattern Match, CodeGen Emitter

NCNN: Develop High Performance CNN Inference Engine *Tencent*, April 2022 - Nov.2022
Mentor: nihui, with 6k+ followers in Github
Optimize high performance neural network operators and math library for NCNN 🏆, 20k+ stars in Github, hand-craftly optimized for x86/ARM/RISCV platforms.

SKILLS

Computing skills: OpenAI Triton, Async Programming, CUDA Graph, SIMD Programming(Arm Neon, X86 AVX512), C/C++, Python
Frameworks: Pytorch, LLVM, MLIR, ColossalAI, Vllm, SGLang, XDiT, Huggingface Transformers
Languages: Native in Mandarin; Fluent in English

Last Update 2024.10.01